

Day 2 — The Magic Constant

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive why 3×3 with 1–9 must sum to 15 per row.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The formula.** For an $n \times n$ magic square using 1 through n^2 , the magic constant is $M = n(n^2 + 1) / 2$.
2. **Verify for 3×3 .** $M = 3 \times 10 / 2 = 15$. ✓
3. **Derive.**
 - Total sum of 1 through $n^2 = n^2(n^2 + 1) / 2$.
 - This sum spreads across n rows, so each row sums to $(n^2(n^2 + 1) / 2) / n = n(n^2 + 1) / 2$.
4. **Predict.** What’s M for 4×4 ? 5×5 ? 7×7 ?
 - 4×4 : $4 \times 17 / 2 = 34$.
 - 5×5 : $5 \times 26 / 2 = 65$.
 - 7×7 : $7 \times 50 / 2 = 175$.
5. **Source.** Nārāyaṇa Paṇḍita’s *Gaṇita-kaumudī* (1356 CE), Book 14, derives the magic constant formula. Earlier references in Tibetan, Chinese, and Arab traditions also give the formula.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Magic Constant.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).